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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

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Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A - all three group members contributed to the project.

1. Data and Deliberately Leaky Baseline

1.1 Security and time-series description

The security chosen is Apple Inc. (AAPL). The daily closing prices were downloaded for the period from January 1 2018 to January 1 2023. The usable return sample spans from January 3 2018 through December 30 2022. The dataset contains 1,258 observations, which's below the project limit of 2,000 observations. This count was obtained after computing percentage returns and removing the first missing return value. The price data were retrieved from Yahoo Finance using the notebook data pipeline (Yahoo Finance).

For the sample range Apple Inc. Closed at prices ranging from \$35.55 to \$182.01. The mean daily return was 0.001101 or 0.11 percent. The standard deviation of returns was 0.021100 or about 2.11 percent. The daily return volatility is much higher than the return, which shows how small the signal is compared to the noise. This low signal-to-noise ratio makes it challenging for any strategy that attempts to forecast returns, for Apple Inc.



Figure 1. AAPL daily closing price over the 2018-2022 sample. The horizontal axis shows date and the vertical axis shows price in U.S. dollars.

1.2 Deliberate leakage design

Two kinds of leakage were purposely added. First the forecasting label is a centered and shifted by one period moving average of five observations. A centered window uses data on both sides of the date which means the label contains future returns. Second the return feature was scaled using the maximum values estimated from the entire sample before the train/test split. Consequently the scaling applied to the training data was tainted with information from the test period. This type of contamination can make historical validation unreliable as López de Prado points out.

The target construction left 1,254 rows. Ten historical returns for each input sequence produced 1,245 sequences. The single split gave 996 sequences for training and 249 sequences, for testing. The global estimation scaling bounds were -0.128647 and 0.119808 .

1.3 Single-split predictive and trading results

We evaluated three deep learning architectures: a multilayer perceptron (MLP), a long short-term memory network (LSTM) and a convolutional network with Gramian Angular Field (GAF) images. LSTM is designed to remember and gate information over sequences (Hochreiter and Schmidhuber). GAF transforms a time series into a two-dimensional angular representation for image-based learning (Wang and Oates). The predictions were translated into a long/short strategy, where a non-negative prediction of the return implied a long position and a negative prediction implied a short position. The strategy return is the position multiplied by the actual return of the next period. Transaction costs were excluded.

Model	MSE	MAE	Directional Accuracy	Cumulative Return	Sharpe	Max Drawdown
MLP	0.000079	0.006965	49.80%	-32.31%	-0.926	-40.19%
LSTM	0.000064	0.006421	47.39%	-29.36%	-0.805	-41.53%
CNN-GAF	0.000527	0.017367	44.18%	-48.79%	-1.726	-55.31%

All three strategies failed to make money in the single test sample, and all directional accuracies were below 50%. The MLP had the best directional accuracy with 49.80% while the CNN-GAF had the worst directional accuracy with 44.18% and the worst loss with -48.79%. The regression errors are measured against the intentionally leaky target, but the directional accuracy and trading returns are measured against the true next-period return. Therefore the table should be seen as a contaminated baseline, not as evidence of real out-of-sample forecasting power.

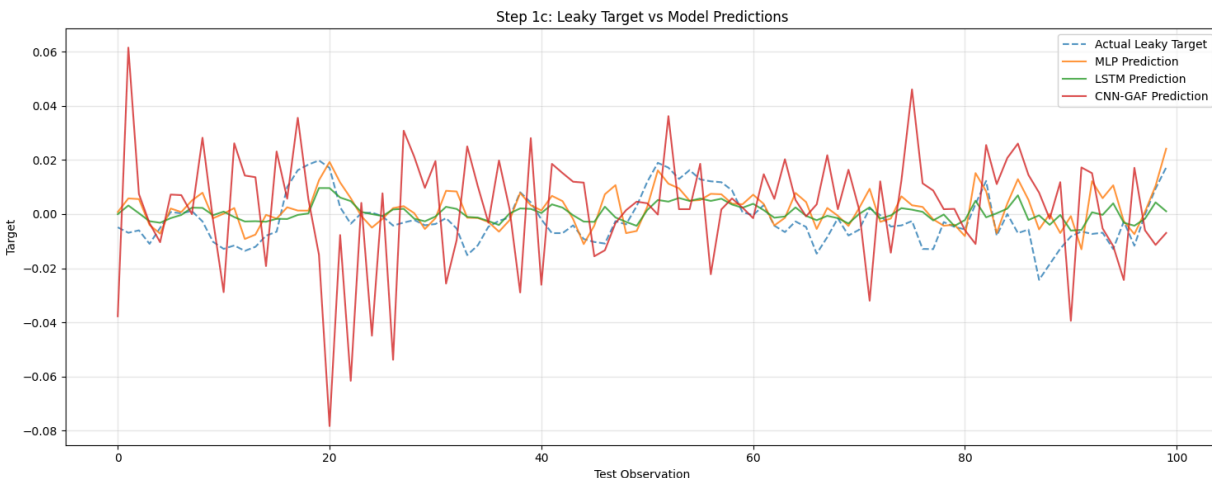


Figure 2. First 100 Step 1 test observations: deliberately leaky target and predictions from the three architectures.

2. Walk-Forward Backtesting with Leakage Retained

We replaced the single split with non-anchored walk-forward validation, but kept the deliberately leaky target and global scaling. The training window was rolling 500 observations per fold. Two test horizons were considered: 500 observations and 100 observations . Models were rebuilt and retrained for each fold. Only the ten observations immediately prior to each test window were used for building the required input sequence context.

2.1 500-training / 500-testing design

Model	Directional Accuracy	Cumulative Return	Sharpe	Max Drawdown
MLP	47.80%	-25.63%	-0.206	-48.55%
LSTM	50.80%	-27.39%	-0.236	-61.66%
CNN-GAF	51.80%	-23.73%	-0.173	-46.04%

Only one complete 500/500 folds could have been formed. Both LSTM and CNN-GAF directional accuracies significantly exceeded 50%, yet each strategy still delivered negative total returns. The CNN-GAF was associated with the least negative cumulative return (-23.73%) of all long-short strategies, whereas the LSTM carry forward the deepest drawdown (-61.66%) – a demonstration that slightly positive or even moderately positive directional accuracy figures are not sufficient for profitable trading due to the interplay of the size and arrangement of gains and losses.

2.2 500-training / 100-testing design

Model	Avg. Directional Accuracy	Avg. Cumulative Return	Avg. Sharpe	Avg. Max Drawdown
MLP	47.29%	-5.33%	-0.287	-24.57%
LSTM	49.43%	-4.93%	0.056	-25.75%
CNN-GAF	53.29%	10.03%	1.079	-18.29%

Seven non-overlapping 100-observation test periods could have been formed. Averaged results were considerably different across folds. Whereas the CNN-GAF achieved 53.29% average directional accuracy and +10.03% cumulative strategy return, other strategies were negative on average. Nevertheless, the fold-level CNN-GAF returns varied from -12.22%, for the first fold, to +31.07% in the fourth fold, which renders a positive average insufficient evidence for consistent profitability.

2.3 Comparison with the single split and between walk-forward horizons

The validation design substantially changed the measured results. Relative to Step 1, the 500/500 walk-forward test improved directional accuracy for the LSTM and CNN-GAF but did not yield a profitable strategy. Shifting to seven 500/100 folds produced a positive average result for CNN-GAF while the other two strategies stayed negative. The shorter test horizon exposes the regime-to-regime variation that a single long test window obscures.

Since leakage is retained in both Step 2 designs, the observed performance cannot be interpreted as clean evidence of generalization. At the same time, the results do not support the claim that leakage underlies every poor or favorable result: with deliberate leakage, many folds and strategies remain negative. A proper conclusion is that validation horizon, historical period, architecture, and leakage jointly influence the apparent backtest outcome.

3. Leakage-Reduced Walk-Forward Validation

3.1 Leakage mitigation

The target was replaced with the strictly forward-looking next-period return, $\text{Target}(t) = \text{Return}(t+1)$. Furthermore, scaling parameters were estimated separately within each 500-observation training window and applied to its associated test window. Thus, test observations no longer dictate the feature transformation used to train earlier observations. This maintains the chronological separation between the training information and future evaluation data, per the leakage-aware financial backtesting practice (López de Prado).

As an implementation audit, the clean target correlates 1.0000 with the actual next-period return since they are intentionally the same quantity. The earlier leaky target correlates only 0.3906 with the actual next-period return. The 1.0000 value validates the label alignment; it is not a measure of predictive performance.

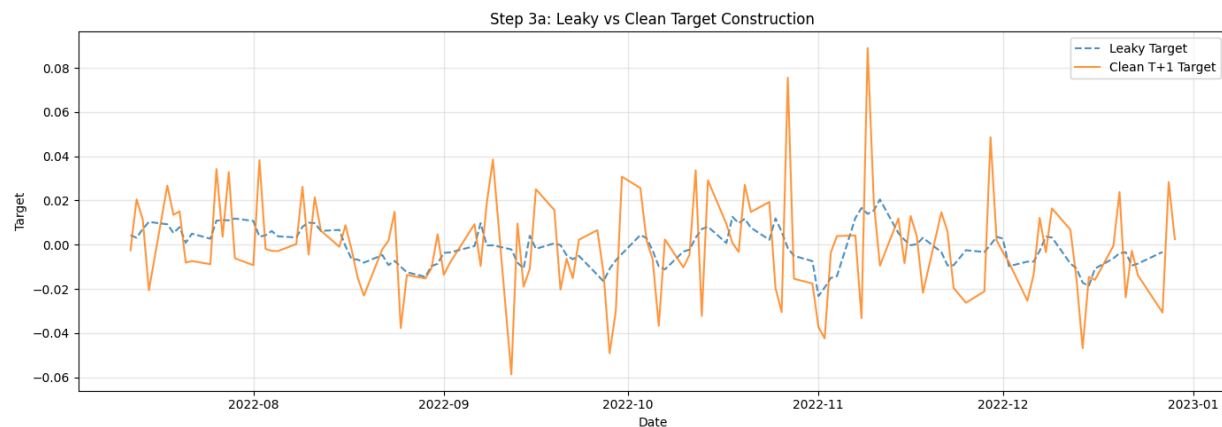


Figure 3. Comparison of the deliberately leaky target and the clean next-period-return target over the final 120 observations.

3.2 Clean 500-training / 500-testing design

Model	Directional Accuracy	Cumulative Return	Sharpe	Max Drawdown
MLP	48.80%	-29.11%	-0.269	-51.31%
LSTM	45.40%	-65.57%	-1.237	-67.66%
CNN-GAF	52.20%	-10.30%	0.043	-46.31%

With leakage controls applied, the single 500/500 fold still did not generate a strong trading result. The CNN-GAF retained directional accuracy slightly above 50% and improved its cumulative return from -23.73% in the leaky design to -10.30% in the clean design. In contrast, the LSTM deteriorated sharply to -65.57%. Since only one 500/500 fold exists, such changes are highly period-dependent and should not be generalized.

3.3 Clean 500-training / 100-testing design

Model	Avg. Directional Accuracy	Avg. Cumulative Return	Avg. Sharpe	Avg. Max Drawdown
MLP	50.14%	-5.12%	-0.177	-25.48%
LSTM	48.43%	-5.42%	-0.057	-22.76%
CNN-GAF	50.86%	2.23%	0.269	-19.43%

Across seven clean 100-observation test folds, all the average directional accuracies were close to the 50% reference level. The CNN-GAF produced a tiny positive average cumulative return of 2.23% but its average Sharpe ratio dropped to 0.269. The MLP and LSTM were still slightly negative. These results offer little evidence of a large stable directional forecasting advantage following chronological leakage controls.

3.4 Effect of leakage mitigation and assessment of apparent overfitting

Model	Leaky 500/100 Accuracy	Clean 500/100 Accuracy	Change	Leaky Return	Clean Return
CNN-GAF	53.29%	50.86%	-2.43 pp	10.03%	2.23%
LSTM	49.43%	48.43%	-1.00 pp	-4.93%	-5.42%
MLP	47.29%	50.14%	+2.86 pp	-5.33%	-5.12%

Leakage mitigation does not affect all architectures in the same direction. For the CNN-GAF, directional accuracy drops by 2.43 percentage points and average cumulative return falls from 10.03% to 2.23%, consistent with leakage inflating its apparent performance. The LSTM loses 1.00 percentage point of directional accuracy. The MLP, however, improves from 47.29% to 50.14% and its average return gets slightly less negative. Thus, the experiment is not a basis to claim a uniform inflation of every model.

The 500/500 comparison echoes this point: after cleaning, CNN-GAF return improves from -23.73% to -10.30% while the LSTM deteriorates from -27.39% to -65.57%. Apparent overfitting has therefore not simply "disappeared". Rather, the exercise demonstrates that performance estimates are sensitive to leakage controls and test-window construction and that robustness should be evaluated over multiple chronologically valid out-of-sample periods rather than one favorable backtest.

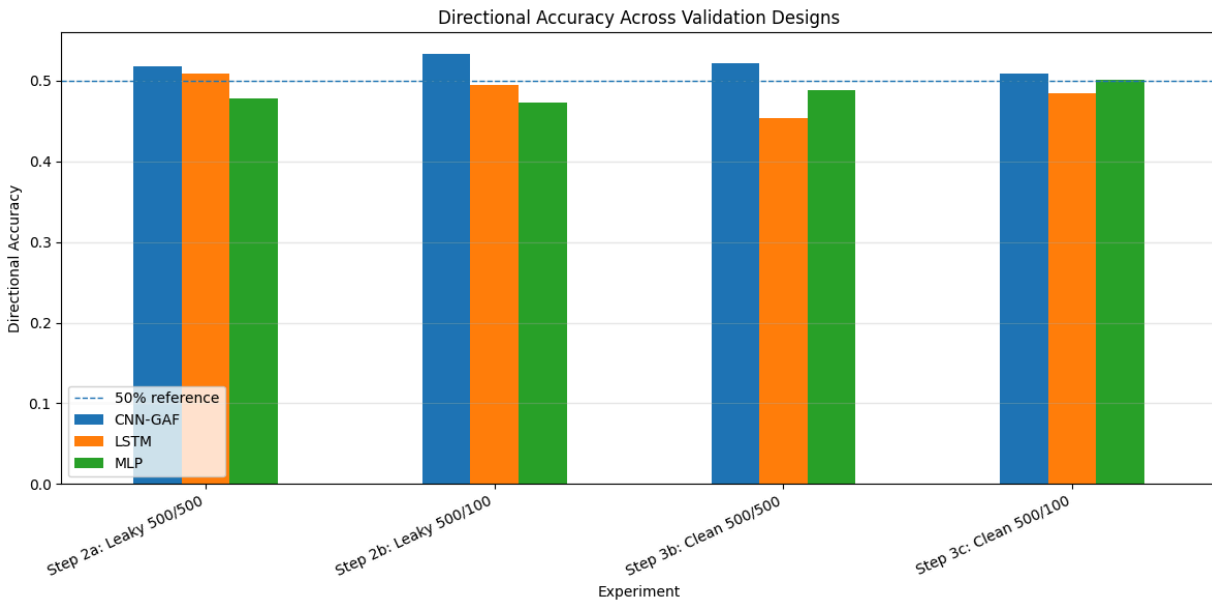


Figure 4. Directional accuracy across the four walk-forward validation designs. The dashed line marks the 50% reference level.

4. Practical Investment Takeaways

The central investment lesson is that the attractive backtest statistics cannot be blindly accepted without examining how the training and test samples were constructed. In this experiment, the change of the validation protocol materially altered the measured accuracy, cumulative return, Sharpe ratio, and drawdown. A strategy that is favorably in one set of 100-observation windows can turn unprofitable in a longer window or after leakage controls are imposed.

For deployment, the evidence supports the use of leakage-aware walk-forward evaluation with preprocessing estimated only from historical training data, multiple out-of-sample windows, and joint review of predictive and economic metrics. Directional accuracy alone is not sufficient: several tests show accuracy around or above 50% but cumulative return remaining negative. Transaction costs, slippage, turnover, and additional market regimes were not modeled, so the reported trading results should be interpreted as validation diagnostics rather than a deployable investment recommendation.

5. Conclusion

Using 1,258 daily AAPL observations, the project intentionally created a contaminated baseline and then progressively strengthened the validation design. The results show that information leakage and test-window choice can materially change performance, but the effect is not uniform across architectures or sufficient to explain every instance of good or bad performance. The clean seven-fold 500/100 experiment leaves all three models close to chance-level directional accuracy, with only a tiny positive average return for the CNN-GAF. The strongest conclusion is therefore methodological: credible financial model assessment requires chronological separation of information, fold-specific preprocessing, repeated walk-forward testing, and evaluation of both prediction quality and realized strategy risk-return outcomes.

References

- Hochreiter, Sepp, and Jürgen Schmidhuber. "Long Short-Term Memory." *Neural Computation*, vol. 9, no. 8, 1997, pp. 1735-1780. doi:10.1162/neco.1997.9.8.1735.
- López de Prado, Marcos. *Advances in Financial Machine Learning*. Wiley, 2018.
- Wang, Zhiguang, and Tim Oates. "Imaging Time-Series to Improve Classification and Imputation." *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence*, 2015, pp. 3939-3945.
- WorldQuant University. "MScFE 642 Deep Learning for Finance: Group Work Project #2." Revised 15 May 2026, pp. 1-4.
- Yahoo Finance. "Apple Inc. (AAPL) Historical Data." Yahoo Finance, finance.yahoo.com/quote/AAPL/history. Data accessed through the notebook data pipeline.